

César Sabater - Curriculum Vitae



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Research Experience

March 2023 **Postdoctoral Researcher**
- present *Institution:* CNRS, INSA Lyon
Topic: Privacy and Security of Data-Processing Systems
Supervisor: Sonia Ben Mokhtar
Research Team: DRIM

July 2022 **Postdoctoral Researcher**
- March 2023 *Institution:* INRIA Lille - Nord Europe
Topic: Privacy-preserving machine learning
Supervisor: Jan Ramon
Research Team: MAGNET

Jan 2019 **Ph.D. in Computer Science**
- June 2022 *Institution:* INRIA Lille - Nord Europe
Defense date: June 20, 2022
Topic: Privacy-preserving machine learning
Supervisor: Jan Ramon
Research Team: MAGNET

Publications

Conferences and Journals

- (SBR 2025) **Dropout-Robust Mechanisms for Differentially Private and Fully Decentralized Mean Estimation**
César Sabater, Sonia Ben Mokhtar, and Jan Ramon
To appear in Proceedings on Privacy Enhancing Technologies (PoPETs) 2026
URL: <https://hal.science/hal-05308749>
- (NSM+ 2025) **PriviRec: Confidential and Decentralized Graph Filtering for Recommender Systems**
Julien Nicolas, César Sabater, Mohamed Maouche, Mark Coates, and Sonia Ben Mokhtar
ACM International Conference on Information and Knowledge Management (CIKM), 2025
URL: <https://hal.science/hal-05308553/>
- (TJB+ 2025) **Exposing the Vulnerability of Decentralized Learning to Membership Inference Attacks Through the Lens of Graph Mixing**
Ousmane Touat, Jezekael Brunon, Yacine Belal, Julien Nicolas, César Sabater, Mohamed Maouche, and Sonia Ben Mokhtar
ACM Middleware Conference, 2025
URL: <https://hal.science/hal-04933985v2>
- (SHP+ 2023) **Private Sampling with Identifiable Cheaters**
César Sabater, Florian Hahn, Andreas Peter, and Jan Ramon
Proceedings on Privacy Enhancing Technologies (PoPETs) 2023
URL: <https://inria.hal.science/hal-03904200/>

(SBR 2022) **An Accurate, Scalable and Verifiable Protocol for Federated Differentially Private Averaging**
César Sabater, Aurélien Bellet, and Jan Ramon
Machine Learning Journal, 2022
URL: <https://inria.hal.science/hal-03820603>

Preprints

(NSM+ 2024) **Differentially Private and Decentralized Randomized Power Method**
Julien Nicolas, César Sabater, Mohamed Maouche, Sonia Ben Mokhtar, and Mark Coates
arXiv preprint, 2024
URL: <https://arxiv.org/pdf/2411.01931>

International Workshops

Zero Knowledge Arguments for Verifiable Sampling
César Sabater and Jan Ramon.
NeurIPS 2021 Workshop on Privacy in Machine Learning, 2021.
URL: <https://openreview.net/pdf?id=YS47RZFtKm>

Distributed Differentially Private Averaging with Improved Utility and Robustness to Malicious Parties
César Sabater, Aurélien Bellet, and Jan Ramon.
NeurIPS 2020 Workshop on Privacy Preserving Machine Learning (PriML & PPML Joint Edition) 2020
URL: <https://arxiv.org/abs/2006.07218>

Reviewing Service

- ICML 2023, 2024, 2025: Reviewer
- NeurIPS 2023, 2024, 2025: Reviewer. For Neurips 2023 I was ranked by area chairs among the top 8% of the reviewers.
- ICLR 2024, 2025: Reviewer
- AAAI 2024: PC Member
- ICDCS 2024: Reviewer
- ACM TOPS 2025: Reviewer
- IH&MMSec Workshop 2024: PC Member

Dissemination in Scientific Venues

July 7-11 2024 **Keynote Speaker at CYBER IN Occitanie 2025 Summer School in Cybersecurity**
Font Romeu, France
<https://cyberoccitanie.sciencesconf.org/>

June 24-27 2024 **Oral Presentation and Session Chair at 14ème Atelier sur la Protection de la Vie Privée (APVP 2024)**
14th Privacy Protection Workshop
Lou Capitelle, France
<https://apvp24.sciencesconf.org/>

Jan 29 -Feb 02 2024 **Session Chair at PEPR Cybersecurité Winter School 2024**
Autrans, France
<https://www.pepr-cybersecurite.fr/2023/11/10/ecole-d-hiver-2024/>

- Sep 18-22 2023 **Oral Presentation at the 22th European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD 2023)**
Torino, Italy
<https://ecmlpkdd.org/2023/>
- July 15-20 2023 **Oral presentation at the 24th Privacy Enhancing Technologies Symposium (PETs 2023)**
Lausanne, Switzerland
<https://petsymposium.org/2024/>
- March 9-10 2023 **Speaker at 2nd RedChain-Lab Workshop**
La Clusaz, France
<https://sites.google.com/view/red-chain-lab-laclusaz/>
- Jan 13 2023 **Speaker at Workshop de la chaire Data et Services pour une Ville Durable (DSVD)**
Lyon, France
<https://sites.google.com/view/workshop-dsvd-2023/accueil>
- June 23-24 2022 **Oral presentation at ANR Privacy-Preserving Methods for Medical Research (PMR) Workshop 2022**
Lille, France
<https://project.inria.fr/anrpmr/project-meeting-23-24-06-2022/>
- Dec 14 2021 **Poster and Recording at NeurIPS 2021 workshop on Privacy in Machine Learning (PriML)**
Virtual workshop
<https://priml2021.github.io/>
- Oct 27-28 2021 **Oral presentation at ANR Privacy-Preserving Methods for Medical Research (PMR) Workshop 2021**
Lyon, France
<https://project.inria.fr/anrpmr/project-meeting-27-10-2021/>
- December 11 2020 **Poster and Recording at NeurIPS 2020 workshop on Privacy Preserving Machine Learning**
PriML and PPML Joint Edition
Virtual workshop
<https://ppml-workshop.github.io/ppml20/>
- June 23-26 2020 **Recorded Presentation at cAP 2020 Conference**
Virtual workshop in France
<https://cap-rfiap2020.sciencesconf.org/>
- July 9-11 2019 **Poster Presentation at APVP 2019 — l’Atelier sur la Protection de la Vie Privée**
Cap Hornu, France
<https://project.inria.fr/apvp2019/>

Awards and Distinctions

- 2006 **5th position in National Selection Contest for IOI (International Olympiad in Informatics)**
1st substitute of the Argentinian team for the IOI that took place in Yucatan, México.
Buenos Aires, Argentina

- 2006 **Bronze Medal (3rd position) in Olimpiada Informática Argentina (Argentinian Olympiad in Informatics)**
National Contest
Buenos Aires, Argentina
- 2004 **5th position in Olimpiada Informática Argentina (Argentinian Olympiad in Informatics)**
National Contest
Buenos Aires, Argentina

Education

- Jan - March 2012 **Master Course Scholarship at IMPA (National Institute of Pure and Applied Mathematics)**
Rio de Janeiro, Brazil
Master's-level course in extremal and probabilistic graph theory and combinatorics, taught by [Robert Morris](#).
Grade: A-
- March 2008 **"Licenciado" Degree in Computer Science (Equivalent to a Master's degree in France)**
April 2017 *University: Universidad Nacional de Rosario (National University of Rosario)*
Academic Average: 9.47/10 (top 5% of the class)
Completed alongside several years of professional activity

Pre-Research Professional Experience

- Jan - Dec 2018 **Research Engineer at INRIA Lille — MAGNET Team, France**
Topic: Development of cryptographic and graph-sampling tools
Supervisor: Jan Ramon
- May - Nov 2014 **Internship at INRIA — CAMUS (Compilation for Multicore Architectures) Team, France**
Topic: Combining compiler optimization and numerical analysis for performance and productivity
- Jan 2012 - Oct 2013 **Software Engineer at CIFASIS-CONICET, Argentina**
(International French-Argentine Center for Information Sciences and Systems)
Topic: Automatic metadata extraction and streamlined submission workflows for open-access academic repositories
- July - Dec 2011 **Software Engineer at ITX Corp**
Co-founder and member of an independent Argentinian team of freelance developers working on assignments for ITX Corp., NY, USA
Main Tasks: Development of front-end mobile websites
- July 2010 - Jun 2011 **Software Engineer at National University of Rosario, Argentina**
Main Tasks: Development of back-end and front-end extensions for online course platforms

Engineering and Programming Skills

Advanced Knowledge in C, C++, STL, Analysis and Algorithm Design, and Data Structures. Advanced Knowledge in Java, PHP and Python. Advanced Knowledge Functional Programming in Haskell and OCaml.

Languages

- French** Advanced writing, reading, oral comprehension and speaking levels.
- English** Advanced writing, reading, oral comprehension and speaking levels.
- Spanish** Native Speaker.